

Abdullah Al Maruf

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Research Summary

I study visual grounding in vision-language models and representation learning for compact vision systems. My work examines how supervision shapes the use of visual evidence in multimodal reasoning. I use evaluation to identify failures that aggregate performance can hide and examine model behavior as data conditions change. I am also interested in knowledge distillation, particularly the tradeoffs between predictive accuracy and computational cost.

Education

Rajshahi University of Engineering & Technology (RUET), Bangladesh

B.Sc. in Computer Science and Engineering

Expected April 2027

CGPA: 3.55/4.00 after six semesters

Research Interests

Visual grounding in vision-language models; multimodal evaluation and robustness; efficient visual representations; knowledge distillation.

Publications and Manuscripts

Accepted Workshop Contributions

Abdullah Al Maruf. “DINOv2 Feature Distillation for Recurrent Local-Update Image Classifiers.” *LIGHT Workshop at NeurIPS 2026*. Peer-reviewed extended abstract; accepted poster; non-archival. [\[Paper\]](#)

Abdullah Al Maruf, Md. Sajedul Islam, Tanmoy Mridha, and Irfan Hossain Bhuiyan. “BD-HazardVLM: Probing Vision-Language Models for Latent Defensive-Driving Hazards in Bangladesh Road Scenes.” *ECCV 2026 Workshop on Safe and Defensive Autonomous Driving (SDAD)*. Accepted workshop paper and poster; non-archival. [\[Paper\]](#)

Submitted Manuscript

Abdullah Al Maruf and Tirtho Roy. “Single-Pass Uncertainty for Selective Polyp Segmentation under External Dataset Variation.” *ML4H 2026 Symposium*. Submitted; under review.

Research Experience

BD-HazardVLM

2026

First author; study design and leadership

- Designed a 500 image Bangladesh road scene benchmark for hazard category, risk timing, severity, defensive actions, and visible evidence.
- The study compared the three VLMs and results showed that hazard-presence accuracy of 88.0-89.2% was very similar to the majority baseline of 89%. Category macro-F1 ranged from 9.4 to 10.5%.
- Identified failures that are masked by aggregate scores: InternVL3-8B has a severity accuracy of 61.2%, but failed all 58 high severity cases.

DINOv2 Feature Distillation

2026

Sole-author study of recurrent local-update image classifiers

- Conducted 12 CIFAR-100 training runs across two architectures, two objectives, and three seeds, with paired initialization, data order, and augmentation views.
- The accuracy of the recurrent-classifier increased from 51.71% to 54.77% (three-seed means), and the feed-forward comparator from 53.99% to 56.13% with the addition of supervision from DINOv2 ViT-S/14.
- Profiled inference costs: the recurrent model required approximately 17.4 times as many MACs as the feed-forward comparator at similar parameter counts.

Research Experience Continued

BanglaNewsGround

Sep 2026 - Present

Undergraduate thesis | Supervisor: Md. Nasif Osman Khansur, Assistant Professor, CSE, RUET

- Comparing journalist captions, text-only rewrites, visually grounded recaptions, and combined supervision to study their effects on Bengali image-text representations.
- Running retrieval and caption-matching experiments with semantic (and entity) hard negatives, to differentiate visual-grounding improvements from language cleanup.

Tiny VLM Relational Reasoning

2026 - Present

Independent research; experiments underway

- Testing question-conditioned structured distillation, answer-only and corrupted-relation controls for sub-billion-parameter VLMs.
- Analyzing counterfactual pairs and visual-dependence controls to assess the sensitivity of predictions to changes in relation and the use of image information.

Spec2Inspect-Lite

2026 - Present

Independent research on industrial inspection; core experiments complete

- Tested a specification-grounded inspection pipeline for three MVTec AD categories and 60 human-written specifications, distinguishing a measurement of the image from the deterministic execution of a rule.
- On 588 matched pairs, the pipeline falsely approved 22 of 288 true FAIL cases (7.64%), versus 272 of 288 (94.44%) for schema-constrained Qwen2.5-VL-3B-Instruct. Coverage was 74.66% versus 99.32%; overall accuracy superiority was not statistically established.
- Evaluated specification-parsing errors on 7,513 specification-image pairs: verdict accuracy was 82.07% with reference rules and 56.73% with raw Qwen-parsed rules.

Visual Reference Tables

2026 - Present

Collaborative research; manuscript in revision

Ushashi Bhattacharjee, **Abdullah Al Maruf**, Koushik Howlader, Sayantan Chakraborty, and Tirtho Roy.

“Visual Reference Tables Recover Signal That Tissue Captions Alone Miss.”

- Continuing work on whether visual reference tables provide useful information beyond tissue captions.

Technical Skills

Programming	Python, C++
Machine Learning	PyTorch, Hugging Face Transformers, PEFT/LoRA, scikit-learn
Research Tools	Git, Docker, Kaggle
Backend and Data	FastAPI, REST APIs, MySQL, PostgreSQL, SQLite

Honors and Awards

- Champion, University Innovation Hub Program, IC-6 cohort.
- Champion, Hult Prize at RUET, 2025; represented RUET at the national competition.
- 6th place, Inter-University Capture the Flag Competition, BUET CSE FEST, 2024.
- Top-25 finalist, BDApps Innovation Challenge, among more than 1,500 teams nationwide.

Leadership and Engineering

GreenLoop | Founder and CTO

Startup producing feed using black soldier fly larvae.

Khoj (Lightning Search)

- Made a fuzzy retrieval system over 10,000+ items with n-gram indexing, candidate filtering, and using RapidFuzz ranking via FastAPI.